

MED-AI Project Overview

Introduction

The **MED-AI** application is a multi-modular healthcare system. It integrates large language models, image classification using a deep learning- model, and an OCR-based medical report summarization module. All features are presented through a single, professional, and interactive streamlit interface.

The aim of this healthcare system is to promote awareness and assist users through accurate AI-powered insights while maintaining ethical safeguards such as medical disclaimers and avoiding hallucinated information.

Modules in MED-AI

1. Medical Chatbot (RAG + LLM)

This intelligent chatbot module uses a Retrieval-Augmented Generation (RAG) pipeline to handle two types of Skin related queries:

- **Symptom-based Queries:** When a user enters symptoms (e.g., "itching and redness"), the system retrieves matching diseases from the FAISS vector store using LangChain retrievers. Users are prompted to select one of the retrieved diseases to receive full details.
- **Disease-specific Queries:** When a user directly asks about a disease (e.g., "What is psoriasis?"), the system retrieves relevant content chunks from the vector store and sends them to the LLM for a structured explanation.

Key Technologies:

- **FAISS VectorStore** for document chunk indexing and retrieval
- **Hugging Face Embeddings** (sentence-transformers/all-MiniLM-L6-v2) to embed document chunks
- **LangChain** for RAG orchestration and query handling
- **Gemma 2 9B** via **Groq API** for context-aware response generation .

2. Skin Disease Image Classifier

This module is designed to classify dermatological conditions from user-uploaded images using a fine-tuned **YOLOv11-small (YOLOv11s)** deep learning model.

The model was trained on a custom dataset sourced from **Roboflow**, which contains a total of **4,773 images** categorized into **three classes**:

- **BCC** (Basal Cell Carcinoma)
- **Atopic Dermatitis**
- **Acne**

The dataset is split as follows:

- **Training Set:** 4,773 images
- **Validation Set:** 210 images
- **Test Set:** 205 images

This dataset was used to fine-tune multiple deep learning classifiers, including **YOLOv11s**, **ResNet-50**, **DenseNet-121**, and **VGG-19**. Among these models, **YOLOv11s** achieved the highest test accuracy of **92%**, making it the preferred choice for integration into the MED-AI application.

After training, the YOLOv11s model was evaluated on the test set to generate a complete classification report, including metrics such as **precision**, **recall**, **F1-score**, and **accuracy** for each class. The classifier works by analyzing the uploaded image and predicting the most likely skin disease category, along with a confidence score that indicates how certain the model is about its prediction.

Model	Training Accuracy	Validation Accuracy	Test Accuracy	Precision	Recall	F1-Score	Model Size
YOLOv11s	0.8352	0.8762	0.9286	0.9288	0.9286	0.9286	10.5 MB
ResNet50	0.5887	0.7095	0.6829	0.7212	0.6829	0.6871	182.6 MB
DenseNet121	0.9172	0.8857	0.8439	0.8428	0.8439	0.8429	42.6 MB
VGG19	0.5014	0.5476	0.4976	0.435	0.4976	0.4124	235.3MB

Key Technologies:

- **YOLOv11s** (You Only Look Once v11 small)
- **Ultralytics** for model training and prediction
- **Pandas & Matplotlib** for generating reports

3. Medical Report Summarizer (OCR + HuggingFace Summarization)

This module allows users to upload scanned or printed medical reports (in image or PDF format), and returns a structured summary using Hugging Face's summarization pipeline.

- If the input is an **image file**, text is extracted using **Tesseract OCR**.
- If the input is a **PDF file**, text is extracted using **PyPDF**
- The extracted text is then summarized using the Hugging Face pipeline
- The default summarization model used is `sshleifer/distilbart-cnn-12-6`

Test Cases : Chatbot

```
Microsoft Windows [Version 10.0.19045.6093]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vr\lin\OneDrive\Desktop\MedAI>C:/Users/vr\lin/anaconda3/Scripts/activate

(base) C:\Users\vr\lin\OneDrive\Desktop\MedAI>conda activate medai_env

(medai_env) C:\Users\vr\lin\OneDrive\Desktop\MedAI>python -m Chatbot.retrievalqa
Welcome to the Medical Assistant Chatbot for Skin Conditions!
Type 'exit' or 'quit' to stop.

You: What is chickenpox?

Bot: Chickenpox is a highly contagious viral infection that causes an itchy rash and blisters, mainly in children. Chickenpox is also known as varicella. The blisters clear up within one to three weeks but may leave a few scars. These are most often depressed (anetoderma), but they may be thickened (hypertrophic scars). Scarring is prominent when the lesions get infected with bacteria. Chickenpox is usually more severe in adults and can be life-threatening in complicated cases.

Retrieved Source Documents:
Document 1:
  Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
  Content: Chickenpox Author: Vanessa Ngan, Staff Writer, 2002. Updated by Dr Jannet Gomez, October 2016; Dr Adam Dedat, SHO and Dr Ian Coulson, Dermatologist , United Kingdom, July 2022. What is chickenpox? Chickenpox is a highly contagious viral that causes an and blistered , mainly in children. Chickenpox is al so known as varicella. The name may be derived from the French term for chickpea, chiche pois. Another theory is that the word 'chicken' was derived from a sla ng term for 'child'. Skin rash of chickenpox Who is at risk of chickenpox? Chickenpox occurs worldwide, affecting persons of all races, and age. Most cases oc cur in children before they are ten years of age. INFECTIONS RASHES infection acute fever rash Chickenpox on back Chickenpox Chickenpox sex
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Document 2:
  Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
  Content: symptoms, vomiting and diarrhoea. Chickenpox is usually more severe in adults and can be life-threatening in complicated cases. Most adults who get chickenpox experience symptoms for up to 48 hours before breaking out in the rash. These include fever, , headache, loss of appetite and abdominal pain. Chick enpox is usually more severe in adults and can be life-threatening in complicated cases. The blisters clear up within one to three weeks but may leave a few sc ars. These are most often depressed (anetoderma), but they may be thickened (hypertrophic scars). Scarring is prominent when the get infected with Cutaneous features of chickenpox susceptible primary papules vesicles prodromal malaise lesions bacteria. Chickenpox on face Chickenpox
```

Document 3:

Source: C:\Users\vrlln\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: fluid from the open sores. A person who is not immune to the virus has a 70-80% chance of being infected with the virus if exposed to someone in the early stages of the disease. What are the clinical features of chickenpox? In children, chickenpox usually begins as itchy red progressing to on the stomach, back and face, and then spreading to other parts of the body. Blisters can also arise inside the mouth The spread pattern can vary from child to child. There may be only a scattering of vesicles, or the entire body may be covered with up to 500 vesicles. The vesicles tend to be very itchy and uncomfortable. Some children may also experience additional symptoms such as high fever, headache, cold-like symptoms, vomiting and diarrhoea.

Document 4:

Source: C:\Users\vrlln\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Once a person has had the chickenpox infection, it is unlikely he or she will get it again, as it confers lifelong immunity. Immunocompromised individuals are to the virus at all times and should take measures to prevent or modify the course of the disease if there has been exposure to the virus. What is the cause of chickenpox? Chickenpox is caused by infection with the varicella-zoster virus, of the Herpesviridae family. This virus is sometimes called herpesvirus type 3. Chickenpox is highly contagious and is easily spread from person to person by breathing in airborne respiratory droplets from an infected person's coughing or sneezing or through direct contact with the fluid from the open sores.

Document 5:

Source: C:\Users\vrlln\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Vaccination against chickenpox Vaccination is available for chickenpox and is highly recommended. Chickenpox is highly preventable by vaccination with live attenuated varicella vaccine. The vaccine is subsidised ("scheduled") for infants aged 15 months in New Zealand as well as non-immune individuals who are immunosuppressed or are in other special groups. If in New Zealand, refer to the Immunisation Advisory Centre for up-to-date information. Trim children's fingernails to minimise scratching. Take a warm bath and apply moisturising cream. Paracetamol can reduce fever and pain Avoid use outside of hospital settings due to the increased risk of severe cutaneous complications such as group A streptococcal superinfections. NSAID invasive

You: treatment

Bot: For most healthy patients with chickenpox symptomatic therapy is usually all that is required.

Retrieved Source Documents:

Document 1:

Source: C:\Users\vrlln\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: For most healthy patients with chickenpox symptomatic therapy is usually all that is required. Immunocompromised patients with chickenpox need intravenous treatment with the antiviral aciclovir. In cases of inadvertent exposure to the virus, varicella-zoster immune globulin if given within 96 hours of initial contact can reduce the severity of the disease though not prevent it. This is used where there is no previous history of chickenpox (or the patient has no to the varicella-zoster virus on blood testing) in pregnancy, in the first 28 days after delivery, and in immune deficient or immune-suppressed patients. How to prevent the spread of chickenpox A person with chickenpox is contagious 1-2 days before the rash appears and until all the blisters have

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Document 3:

Source: C:\Users\vrlln\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Complications may include: Chickenpox Chickenpox Chickenpox incubation detects the varicella virus in skin lesions and is the most accurate method for diagnosis.PCR The use of blister fluid is time-consuming and is less frequently performed.culture (and) is most useful in pregnant women, or before prescribing medication to determine the need for pre-treatment immunisation. Serology IgM IgG immune suppression self-limiting Secondary infection of skin lesions caused by scratchingbacterial Infection may lead to , cellulitis, fasciitis and abscess necrotising gangrene Dehydration from vomiting and diarrhoea of asthmaExacerbation Viral pneumonia

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You:

You: suggest some antibiotics

Bot: This information is not available in the provided context.

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Document 3:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: did show superiority of doxycycline. Macrolides as the penicillin class represent an alternative when traditional antibiotics cannot be used. TMP/SMX and trimethoprim have also been used for the treatment of acne. Sulfamethoxazole is bacteriostatic by blocking bacterial synthesis of folic acid, which is necessary for cell division. Trimethoprim is a folic acid analog that inhibits the enzyme dihydrofolate reductase. The 2 agents work together to block nucleotide and amino acid synthesis in the bacteria. Outside of case reports, there is 1 small, double-blind study showing that TMP/SMX is as effective as oxytetracycline. Although data supporting their use are limited, penicillins and cephalosporins are sometimes used in

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You: i have redness and itching on my face

Bot: Possible conditions include:

- * Seborrheic dermatitis
- * Blepharitis
- * Contact dermatitis

Retrieved Source Documents:

Document 1:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Infantile seborrheic dermatitis - note eczema in the napkin area and and cradle cap axillae Thick yellow scale in cradle cap Inflammatory infantile seborrheic dermatitis - note in the body folds lesions retroauricular Winter flares, improving in summer following sun exposure Minimal itch most of the time Combination oily and dry mid-facial skin Ill-defined scaly patches or diffuse scale in the scalplocalised Blepharitis: scaly red eyelid margins Salmon-pink, thin, scaly, and ill-defined in skin folds on both sides of the face Papules Petal or ring-shaped flaky patches on the hairline and on chestanterior Rash in the armpits, under the breasts, in the groin folds, and creasesgenital Malassezia folliculitis (inflamed hair) on the cheeks and upper trunk.follicles

Document 2:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: The rash involves the eyes, mouth, face or genitals Seek immediate medical care in the following situations: You think your skin is infected. Clues include fever and pus oozing from blisters. It's hard to breathe after inhaling burning weeds. Your eyes or nasal passages hurt after inhaling smoke from burning poison ivy. You think an ingested substance has damaged the lining of your mouth or digestive tract. Causes Contact dermatitis is caused by exposure to a substance that irritates your skin or triggers an allergic reaction. The substance could be one of thousands Request an appointment From Mayo Clinic to your inbox Sign up for free and stay up to date on research advancements, health tips, current health topics, and expertise on managing health. Click here for an email preview.

Document 3:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Is rosacea causing your red face? Rosacea can make your face red all (or most of) the time, but, it's not the only sign. Do you have any of these other signs or symptoms? 4/22/25, 9:27 AM What is rosacea? <https://www.aad.org/public/diseases/rosacea/what-is> 4/4

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Document 4:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: What are the types of dermatitis? dermatitis is the result of an external factor or insult that induces skin inflammation. Common causes include: and fine on the cheeks in alba Hypopigmentation scale pityriasis Cheek and nasolabial fold redness and in dermatitis scaling seborrhoeic Atopic dermatitis and pityriasis alba are more common in children Hand eczema is more common in young and middle-aged adults Venous or gravitational eczema, asteatotic dermatitis, and nummular eczema are more common in middle and older age groups . Acute dermatitis will show redness or swelling, papulation, vesiculation, oozing and weeping, and even blistering. Chronic eczema will show skin thickening with of the skin creases , hyperkeratosis, scaling, fissuring, , and accentuation

Document 5:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Seborrheic dermatitis is a chronic form of eczema that affects oily areas of the body, such as the scalp, face (around the nose and eyebrows) and/or upper and central chest . It causes red, greasy and inflamed skin covered with white or yellowish scales. In adults, it is commonly referred to as dandruff when confined to the scalp, and in infants, it is called "cradle cap." The condition can be recurrent, with flares often triggered by stress, cold weather, hygiene practices or hormonal changes. Treatment often includes medicated shampoos, topical antifungal or corticosteroid creams and maintaining a regular skincare routine to control symptoms. In severe cases, other prescription medications may be necessary. Get more details on symptoms, diagnosis and treatment for seborrheic dermatitis

You: corticosteroid means?

Bot: This information is not available in the provided context.

Retrieved Source Documents:

Document 1:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Systemic corticosteroids or immune suppressive medications Poor circulation or lymphoedema.peripheral asymmetrical unilateral. Itchy and/or between the toes, especially between 4th and 5th toeserosions scales covering the sole and sides of the feet (/moccasin type, usually caused by T. rubrum) Scale hyperkeratotic Small to medium-sized blisters, usually affecting the inner aspect of the foot (type). vesiculobullous ulceration pustules microscopy hyphae culture

Document 2:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: (triiodothyronine) () hyperthyroidism Low High High Secondary hyperthyroidism High High High Subclinical hyperthyroidism Low Normal Normal Primary

Document 3:
Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
Content: Crossref PubMed Google Scholar 249.Rifka, S.M. · Pita, J.C. · Vigersky, R.A. ... Interaction of digitalis and spironolactone with human sex steroid receptors J Clin Endocrinol Metab. 1978; 46:338-344 Crossref PubMed Google Scholar Download PDF

Document 4:
Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
Content: acne, including pulsed dye laser, glycolic acid peels, and salicylic acid peels Intralesional corticosteroid injections are effective in the treatment of individual acne nodules Download PDF

Document 5:
Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
Content: Inflammatory and (papulopustular)papules pustules Phymatous changes Thickening of the skin due to hyperplasia/fibrosis of the sebaceous glands of the face Most common area affected is the nose (termed rhinophyma) More commonly present in men. Rhinophyma Rhinophyma showing swelling and sebaceous openings gland Steroid induced papulopustular rosacea Steroid induced rosacea on the forehead

You: Does eczema spread by touch?

Bot: This information is not available in the provided context.

Retrieved Source Documents:

Document 1:
Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
Content: Symptoms include redness, swelling, itching and sometimes blisters. The reaction is usually limited to the area of skin that comes in contact with the irritant or allergen, but it can be severe if exposure continues. This condition can affect anyone. It can be occupational, meaning it occurs in people exposed to allergens or irritants at work. Once the trigger is identified, avoiding the substance is critical to prevent further reactions. Get more details on symptoms, diagnosis and treatment for contact dermatitis See Photos of Eczema Eczema can vary in appearance for many reasons including type, level of severity, affected part of the body and, notably, skin tone. The Eczema Visual Guide is the largest publicly-available repository of

Document 2:
Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
Content: Gravitational eczema compounded by a contact allergy to a bandage constituent Excoriated acute eczema on the aspects of the knees (reverse pattern) extensor over the knuckles in hand eczema Fissuring atopic dorsal Weepy of eczema on the legs lesions discoid Weepy and impetiginised discoid eczema on the foot Fissuring, and in fingertip dermatitis hyperkeratosis vesiculation chronic Discoid pattern of eczema on the dorsal hands Hand eczema on the back of the hand - mixture of irritant factors in this atopic person Click to show sensitive image

Document 3:
Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
Content: occupational factor; this is erroneous. RASHES TERMINOLOGY Dermatitis inflammatory epidermal Cause eg, allergic contact dermatitis, photosensitive dermatitis Clinical appearance eg, discoid dermatitis, hyperkeratotic dermatitis, and pompholyx Site of eg, hand dermatitis, eyelid dermatitis, or lower leg dermatitis.predisilection irritant endogenous eczema Dermatitis, strictly speaking, includes any cause of skin affecting the inflammation epidermis. Eczema is derived from the Greek word for “to boil or bubble over”, which pathologically manifests as within the epidermis (called).oedema spongiosis mean weepy contact dermatitis due to sticking plaster Acute allergy Allergic contact dermatitis due to nickel in the sides of her spectacle frame

Document 4:
Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
Content: hands, inside the elbows and behind the knees, although it can appear anywhere on the body. Atopic dermatitis often occurs alongside asthma and allergic rhinitis (also known as “hay fever”), and it can run in families. Atopic dermatitis is thought to result from a combination of genetic, environmental and immune system factors. In people with atopic dermatitis, the skin barrier is weakened, making it more susceptible to dryness, irritants, allergens and microbes. Symptoms can be triggered by allergens, harsh soaps, cold weather or stress, among other factors. While it cannot ABOUT ECZEMA MANAGING ECZEMA LIVING WITH ECZEMA RESOURCES RESEARCH GET INVOLVED GET THE LATEST NEWS 4/22/25, 9:30 AM Types of Eczema: Symptoms, Differences, Diagnosis | National Eczema Association

Document 5:
Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
Content: applied liberally and often enough to: Atopic eczema – especially in people with ichthyosis vulgaris Eczema craquelé – especially in older people. Also called asteatotic eczema A dry form of nummular dermatitis/discoid eczema – especially in people that wash their skin excessively. rash pruritus Flaky skin or itchy skin (pruritus) Skin infection when or viruses penetrate a break in the skin surfacebacteria Overheating, especially in some forms of ichthyosis Food allergy, eg, to peanuts, has been associated with filaggrin mutations Contact allergy, eg, to nickel, has also been correlated with barrier function defects. Family history Age of onset Appearance at birth, if known of dry skinDistribution Other features, eg eczema, abnormal , , dentition, sight, hearing.nails hair

You: can albinism be treated with antifungal medication?

Bot: This information is not available in the provided context.

Retrieved Source Documents:

Document 1:
Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf
Content: Other less common types of albinism may also involve problems with blood clotting, immune deficiency or problems with hearing. One concern that should not be overlooked is the risk of isolation in people with albinism. People with albinism, especially children need to be treated normally and included in all activities. They develop normally and have normal intelligence, it is a myth that people with albinism are mentally impaired or intellectually-challenged. What treatment or precautions can be taken? It is important for people with albinism to protect themselves from UV exposure and thus prevent the damaging effects it can have on the skin. Because the patient has no, or little, , skin cancers will often have no or little

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Content: Because the patient has no, or little, , skin cancers will often have no or little pigmentation. Patients with albinism should promptly report suspicious spots or growths to a doctor. Specialist eye doctors cannot cure eye problems but can help with various optical aids to improve vision for people with albinism. : sensitivity to light or glarePhotophobia Nystagmus: involuntary movement of the eyes back and forth : eyes do not fixate and track togetherStrabismus Retinal involvement: this is an important area of the eye as it is responsible for sending signals to the brain. Impaired transmission of signals causes various vision disorders Sun avoidance methods Wear protective clothing (long sleeves and pants, shirts with collars, tightly woven fabrics that

Document 3:

Source: C:\Users\vrllin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: About 1 in 70 people have a gene for albinism. Couples whom are each carriers of the recessive albinism gene have a 1 in 4 chance of producing a child with albinism. What are the problems associated with albinism? The main problems of albinism are caused by the inability of the body to produce melanin pigment (whose major role in the skin is to absorb light from the sun so skin is not sun-damaged). It also has a role in the of normal vision of the eye. Having white or light coloured hair due to lack of melanin is no cause for concern, however, lack of melanin in the skin and eyes can cause the following problems: Skin problems Eye problems Reduced eumelanin synthesis Rufous/red and brown subtypes Similar to OCA 2 Most common type in Japan, China and India Hermansky-Pudlak syndrome

Document 4:

Source: C:\Users\vrllin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Direct treatment may be helped by a handheld ultraviolet B (UVB) comb device. Immunosuppressants such as methotrexate and ciclosporin. Vitamin A derivative acitretin. Phosphodiesterase-4 (PDE4) apremilast.inhibitor Biologic agents such as anti- , IL-17 and IL-12,-23 (eg, etanercept, adalimumab, infliximab, brodalumab, secukinumab, ixekizumab, ustekinumab, guselkumab, risankizumab, and others. TNF inhibitors exacerbating

Document 5:

Source: C:\Users\vrllin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Wash after dirty work with soap and water, rinse off and dry carefully. Apply emollient hand frequently – dimethicone barrier may help.cream creams microbes Topical corticosteroid is applied for 2-4 weeks and repeated for flares.ointment Tacrolimus ointment is an alternative when is not responding to routine management [2]. dermatitis Intralesional steroid injections are sometimes used in resistant cases. Antiseptics or antifungal lotions or solutions may be applied for several months. Oral antifungal agent (itraconazole or fluconazole), if C. albicans is confirmed. Patients with diabetes and disease with toenail paronychia infections should be examined for signs of cellulitis. vascular Surgical of the pro

You: use of Methotrexate

Bot: Methotrexate is a systemic treatment used for psoriasis.

Retrieved Source Documents:

Document 1:

Source: C:\Users\vrllin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: did show superiority of doxycycline. Macrolides as the penicillin class represent an alternative when traditional antibiotics cannot be used. TMP/SMX and trimethoprim have also been used for the treatment of acne. Sulfamethoxazole is bacteriostatic by blocking bacterial synthesis of folic acid, which is necessary for cell division. Trimethoprim is a folic acid analog that inhibits the enzyme dihydrofolate reductase. The 2 agents work together to block nucleotide and amino acid synthesis in the bacteria. Outside of case reports, there is 1 small, double-blind study showing that TMP/SMX is as effective as oxytetracycline. Although data supporting their use are limited, penicillins and cephalosporins are sometimes used in

Document 2:

Source: C:\Users\vrllin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Oral small molecules – baricitinib, upadacitinib, and abrocitinib either are licensed or are being considered for licence and use in moderate/severe atopic dermatitis in many countries. These

Document 3:

Source: C:\Users\vrllin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Commonly used topical acne therapies include BP, salicylic acid, antibiotics, combination antibiotics with BP, retinoids, retinoid with BP, retinoid with antibiotic, azelaic acid, and sulfone agents. Although most physicians have anecdotal regimens they find beneficial, agents reviewed here are limited to those approved by the US Food and Drug Administration (FDA) for use in the United States, and for which peer-reviewed literature has been published. BP is an antibacterial agent that kills P acnes through the release of free oxygen radicals and is also mildly comedolytic. No resistance to this agent has been reported, and the addition of BP to regimens of antibiotic therapy enhances results and may reduce resistance development. BP is

Document 4:

Source: C:\Users\vrllin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Topical calcipotriol Topical calcineurin inhibitors Systemic treatments Methotrexate Acitretin Biological agents for psoriasis – infliximab, adalimumab, etanercept, ustekinumab Novel small – apremilast, tofacitinib [see Janus kinase inhibitors]molecules Nonpharmacological treatments Phototherapy Lasers Photodynamic therapy relapses

Document 5:

Source: C:\Users\vrllin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Retinoid use may be limited by side effects, including dryness, peeling, erythema, and irritation, which can be mitigated by reduced frequency of application. Given any sing

een can be used to reduce the risk of sunburn.

You: what is Erdheim Chester diseases?

Bot: This information is not available in the provided context.

Retrieved Source Documents:

Document 1:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: What are the complications of acute rheumatic fever? The most severe complication of recurrent ARF is permanent damage to heart valves, known as rheumatic heart disease. The disease can result in permanent damage to: How is acute rheumatic fever diagnosed? Erythema marginatum rheumaticum - This is a characteristic type of annular erythema that occurs in about 10% of first attacks of ARF in children; it is very rare in adults. The can be difficult to detect in dark-skinned people. When present, it is found on the trunk and upper arms and legs, but almost never on the face, palms or soles. The rash appears as pink or red (flat spots) or (small lumps), which spread outwards in a circular shape. As the

Document 2:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Hand, foot, and mouth disease August 2022 Author: Dr Stanley Leong, Paediatric Registrar; Dr Caroline Mahon, Dermatologist, Christchurch Hospital, New Zealand Previous contributors: A/Prof Amanda Oakley, Dermatologist (1998 and 2016); Dr Jannet Gomez (2016) Reviewing dermatologist: Dr Ian Coulson Edited by the DermNet content department What is hand, foot, and mouth disease? Hand, foot, and mouth disease (HFMD or HFM) is a common, , viral that causes blisters on the hands, feet, and inside or around the mouth. It mainly affects children under the age of 5 years. HFMD, also called enteroviral , occurs sporadically worldwide. are most common during warm weather, usually in the late summer or early autumn.

Document 3:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: Yellow nail syndrome Author: Vanessa Ngan, Staff Writer, 2006. What is yellow nail syndrome? Yellow is a rare disorder of the nail, which is usually accompanied by lymphoedema (swelling of parts of the body caused by blockage or damage to the drainage of the system). It may also be associated with (fluid collection in space surrounding the lungs) and less commonly bronchiectasis (, abnormal dilation of the bronchi in the lungs), chronic bronchitis and SYSTEMIC DISEASES nailsyndrome lymphatic recurrentpleural effusions chronic sinusinfections. Yellow nail syndrome Yellow nail syndrome Yellow nail syndrome Yellow nail syndrome

Document 4:

Source: C:\Users\vr\lin\OneDrive\Desktop\MedAI\Chatbot\data\SkinDisease.pdf

Content: The skin of acute rheumatic fever is marginatum. Who gets rheumatic fever? ARF usually affects children aged 5-15 years. Most cases of ARF currently occur in countries. Worldwide there is an estimated 470 000 new cases of ARF annually (60% of whom eventually rheumatic heart disease). In most developed countries ARF is now rare, with a few notable exceptions; the highest documented rates of ARF in the world are in Maori and Pacific people in New Zealand, Aboriginal Australians, and those in Pacific Island nations. Rheumatic fever is associated with poverty, overcrowding and poor sanitation facilities. It is suspected that there are that make some families more to the disease. What are the clinical features of acute rheumatic fever?

Skin Disease Classifier output

Predicted: BCC (99.9%)\nTrue: BCC



Predicted: ACNE (100.0%)\\nTrue: ACNE



Predicted: ACNE (84.0%)\\nTrue: ATOPIC



Report Summarizer

Input Image 1

Exam Number: 12345678	Report Status: Final
Type: Chest 2 Views	
Date/Time: 01/01/2014 10:30	
Exam Code: XRCH2	
Ordering Provider: Wayne, John Michael MD	
HISTORY:	
- Cough and Fever	
REPORT	Frontal and lateral views of the chest.
COMPARISON: None	
FINDINGS:	
Lines/tubes: None.	
Lungs: The lungs are well inflated and clear. There is no evidence of pneumonia or pulmonary edema.	
Pleura: There is no pleural effusion or pneumothorax.	
Heart and mediastinum: The cardiomediastinal silhouette is normal.	
Bones: The visualized skeleton is normal.	
IMPRESSION:	
Clear lungs without evidence of pneumonia.	
RECOMMENDATION:	
None.	
PROVIDERS:	SIGNATURES:
Doe, Jane Lynn MD	<u>Doe, Jane Lynn MD</u>
If you have questions or concerns regarding this report, feel free to	

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 XRAY.jpg 102.6KB

×

Summary

The lungs are well inflated and clear. There is no evidence of pneumonia or pulmonary edema . The cardiomediastinal silhouette of the heart and mediastinum is normal .

Name : Mr. Ayush Chauhan
Age/Gender : 23 Years/Male
Referred By : Self
Phone No. : 6397228160

Patient ID : 240318002
Report ID : RE9
Collection Date : 19/03/2024 03:06 PM
Report Date : 19/03/2024 03:07 PM



RADIOLOGY
CT SCAN OF BRAIN (PLAIN STUDY)

Serial axial images of brain were studied without administration of IV contrast.

Observations:

Attenuation values of brain parenchyma normal
No evidence of intra or extra axial hemorrhage or collection noted.
No evidence of mass lesions.
No shift of midline structures.
Ventricular system normal.
Cisternal spaces and cortical sulcal gyral pattern normal.
Posterior fossa structures are normal.
No evidence of bony injury.

Impression: * No significant abnormality detected in brain parenchyma.

For clinical correlation.



~~End of report~~

Dr. Kevin
M.D Radiologist

Dr. John
M.D Radiologist

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CT.png 247.9KB

×

Summary

No significant abnormality detected in brain parenchyma . No evidence of mass lesions or mass lesions . No significant abnormalities detected . No signs of bony injury .